

Direct intervention versus imperfect implementation

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1 Overview

A manager cannot set grazing pressure. What can be set is a policy; whether it lowers the pressure depends on compliance and on the stocking rate that results. SPEC section 42 asks for this chain to be modelled explicitly,

```
ManagementPolicy -> Compliance -> StockingRate -> GrazingPressure -> Vegetation
```

rather than only as `do(GrazingPressure = low)`. This vignette builds the chain as an open network, glues it onto the reference habitat network of SPEC section 45 with the composition operations of `CategoricalBayesianNetworks.jl`, compares the direct intervention with the

implementable one under uncertain compliance, and finishes with the cost-benefit trade-off as a small influence diagram.

Modelling the lever rather than the outcome is what makes the result usable for management: a conservation decision is taken over the policy that can be set, not over the ecological variable one wishes to control (Marcot et al. 2006; Runge et al. 2011).

2 Setup

```
using EcologicalBayesianNetworks
using BayesianNetworks
using CategoricalBayesianNetworks: Open, compose, glue, apex, inputs, outputs
using InfluenceDiagrams
ref = reference_habitat_model()
```

```
BayesModel(7 variables, 7 mechanisms, 7 kernels)
```

3 The implementation chain as an open network

The chain is a closed network on its own (the policy has a prior), and its GrazingPressure is an output. The habitat network is written with GrazingPressure left exogenous (closed = false), so that it can be an input. Open checks the typed-interface rule: inputs have no mechanism, every mechanism-free variable is an input.

```
chain = bayesnet(:ManagementPolicy ⇒ [:restrict, :none], :Compliance ⇒ [:high,
:low],
                :StockingRate ⇒ [:low, :high], :GrazingPressure ⇒ [:low, :high];
mechanisms = [:Compliance ⇒ :ManagementPolicy, :StockingRate ⇒
:Compliance,
                :GrazingPressure ⇒ :StockingRate])
habitat = bayesnet(:Climate ⇒ [:dry, :normal, :wet], :Irrigation ⇒ [:low, :high],
                  :SoilMoisture ⇒ [:low, :medium, :high], :GrazingPressure ⇒
[:low, :high],
                  :Vegetation ⇒ [:sparse, :moderate, :dense], :HabitatQuality ⇒
[:poor, :good],
                  :Occupancy ⇒ [:absent, :present];
mechanisms = [:Climate ⇒ (), :Irrigation ⇒ (),
                :SoilMoisture ⇒ (:Climate, :Irrigation),
                :Vegetation ⇒ (:SoilMoisture, :GrazingPressure),
                :HabitatQuality ⇒ :Vegetation, :Occupancy ⇒
:HabitatQuality],
                closed = false)
C = Open(chain; outputs = [:GrazingPressure])
H = Open(habitat; inputs = [:GrazingPressure], outputs = [:Occupancy])
println(C)
```

```
println(H)
```

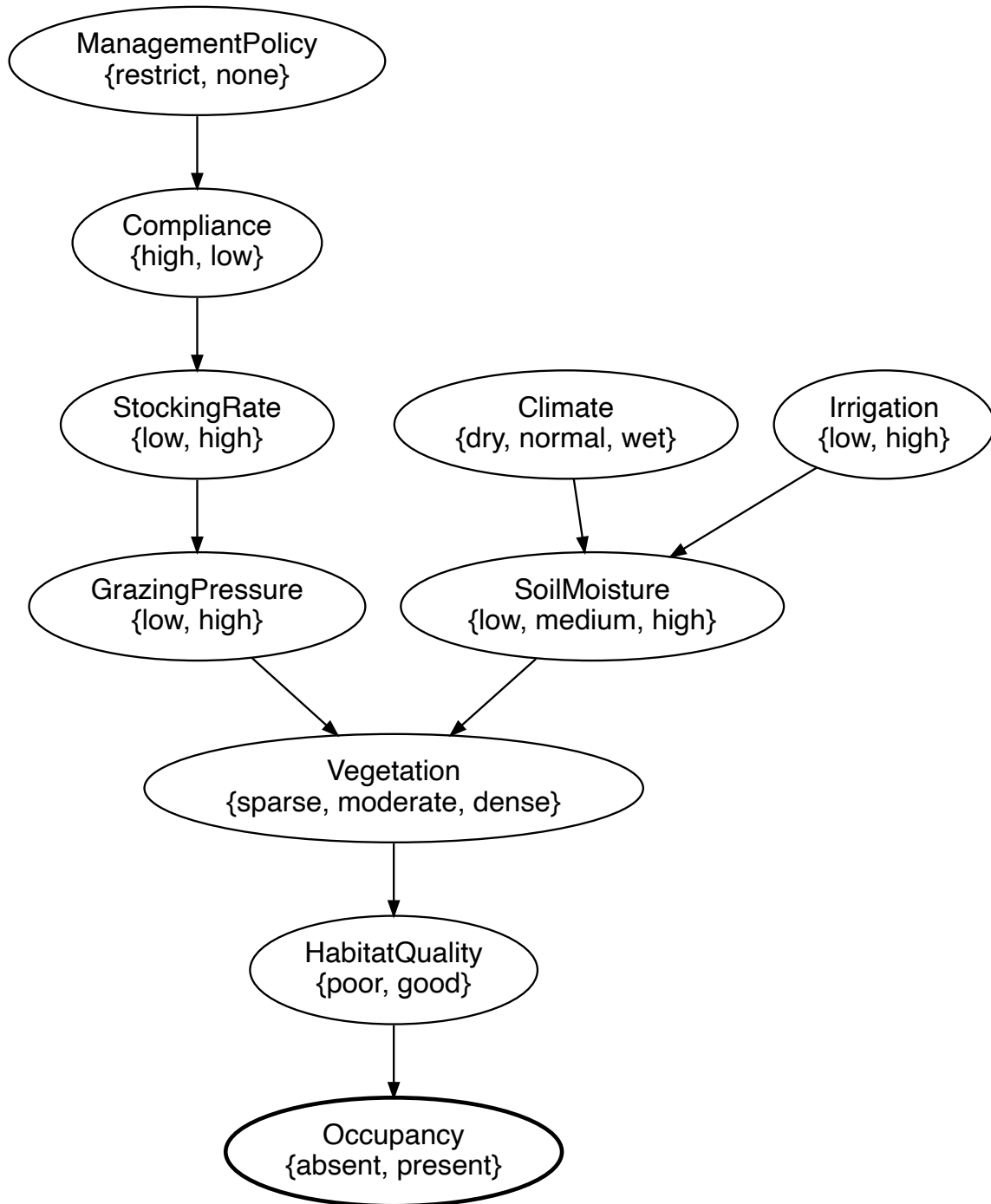
```
OpenBayesNet(4 variables, 4 mechanisms; Symbol[] -> [:GrazingPressure])  
OpenBayesNet(7 variables, 6 mechanisms; [:GrazingPressure] -> [:Occupancy])
```

compose glues the output of the chain to the input of the habitat network (a pushout along the shared variable space); glue(C, H; along = [:GrazingPressure $\Rightarrow$:GrazingPressure]) does the same with explicit pairs. The composite has no inputs left and Occupancy as its output.

```
G = compose(C, H)  
inputs(G), outputs(G), sort(variable_names(apex(G)))
```

```
(Symbol[], [:Occupancy], [:Climate, :Compliance, :GrazingPressure, :HabitatQuality,  
:Irrigation, :ManagementPolicy, :Occupancy, :SoilMoisture, :StockingRate,  
:Vegetation])
```

```
to_graphviz(G)
```



4 Binding the numbers

The apex of the composite is an ordinary closed network. The habitat kernels are copied from the reference model (`bind_kernel` accepts a kernel whose axes match the mechanism's parents);

the chain gets tables of its own: a restriction is complied with 70% of the time, compliance keeps the stocking rate low 90% of the time, and a low stocking rate gives low pressure 90% of the time.

```
full = BayesModel(apex(G))
full = bind_kernel(full, [x ⇒ kernel(ref, x)
                          for x in (:Climate, :Irrigation, :SoilMoisture,
:Vegetation,
                                :HabitatQuality, :Occupancy)])
full = bind_cpt(full, [:ManagementPolicy ⇒ [0.5, 0.5],
                    :Compliance ⇒ [0.7 0.3; 0.1 0.9],      # rows: restrict,
                    none
                    :StockingRate ⇒ [0.9 0.1; 0.2 0.8],      # rows: high, low
                    compliance
                    :GrazingPressure ⇒ [0.9 0.1; 0.15 0.85]]) # rows: low, high
                    stocking
validate(full; closed = true, unique_names = true, semantics = true)
```

5 Direct intervention versus deciding the policy

`do(GrazingPressure = low)` is the ideal: the pressure is low with certainty. `do(ManagementPolicy = restrict)` is what a manager can do; the pressure is then low only with the probability the chain allows.

```
present(m) = round(marginal(m, :Occupancy).table[2]; digits = 4)
(baseline = present(full),
 do_pressure_low = present(do_intervention(full, :GrazingPressure ⇒ :low)),
 do_policy_restrict = present(do_intervention(full, :ManagementPolicy ⇒
:restrict)),
 do_policy_none = present(do_intervention(full, :ManagementPolicy ⇒ :none)))
```

```
(baseline = 0.4769, do_pressure_low = 0.5117, do_policy_restrict = 0.4881,
 do_policy_none = 0.4658)
```

```
round.(marginal(do_intervention(full, :ManagementPolicy ⇒ :restrict),
:GrazingPressure).table; digits = 4)
```

```
2-element Vector{Float64}:
 0.6675
 0.3325
```

Restricting grazing moves the occupancy probability only part of the way towards what guaranteed low pressure would give, because a third of the time the restriction does not translate into low pressure. Since `ManagementPolicy` is a root, observing the policy and intervening on it agree; the difference between `observe` and `do_intervention` appears further down the chain, where observing low pressure is also evidence about compliance and the policy:

```
(observe_low = round.(marginal(observe(full, :GrazingPressure => :low),
:ManagementPolicy).table; digits = 4),
do_low = round.(marginal(do_intervention(full, :GrazingPressure => :low),
:ManagementPolicy).table; digits = 4))
```

```
(observe_low = [0.6544, 0.3456], do_low = [0.5, 0.5])
```

6 Compliance as the lever

Rebinding the Compliance table sweeps the probability that a restriction is complied with; the effect of the policy on occupancy grows with it and reaches the direct intervention only in the limit of perfect implementation.

```
println(" P(comply)   P(present | do(restrict))")
for p in (0.3, 0.5, 0.7, 0.9, 1.0)
    m = bind_cpt(full, :Compliance => [p 1-p; 0.1 0.9])
    println(lpad(p, 8), "      ", present(do_intervention(m, :ManagementPolicy =>
:restrict)))
end
```

P(comply)	P(present do(restrict))
0.3	0.4732
0.5	0.4807
0.7	0.4881
0.9	0.4955
1.0	0.4993

7 The cost-benefit trade-off as an influence diagram

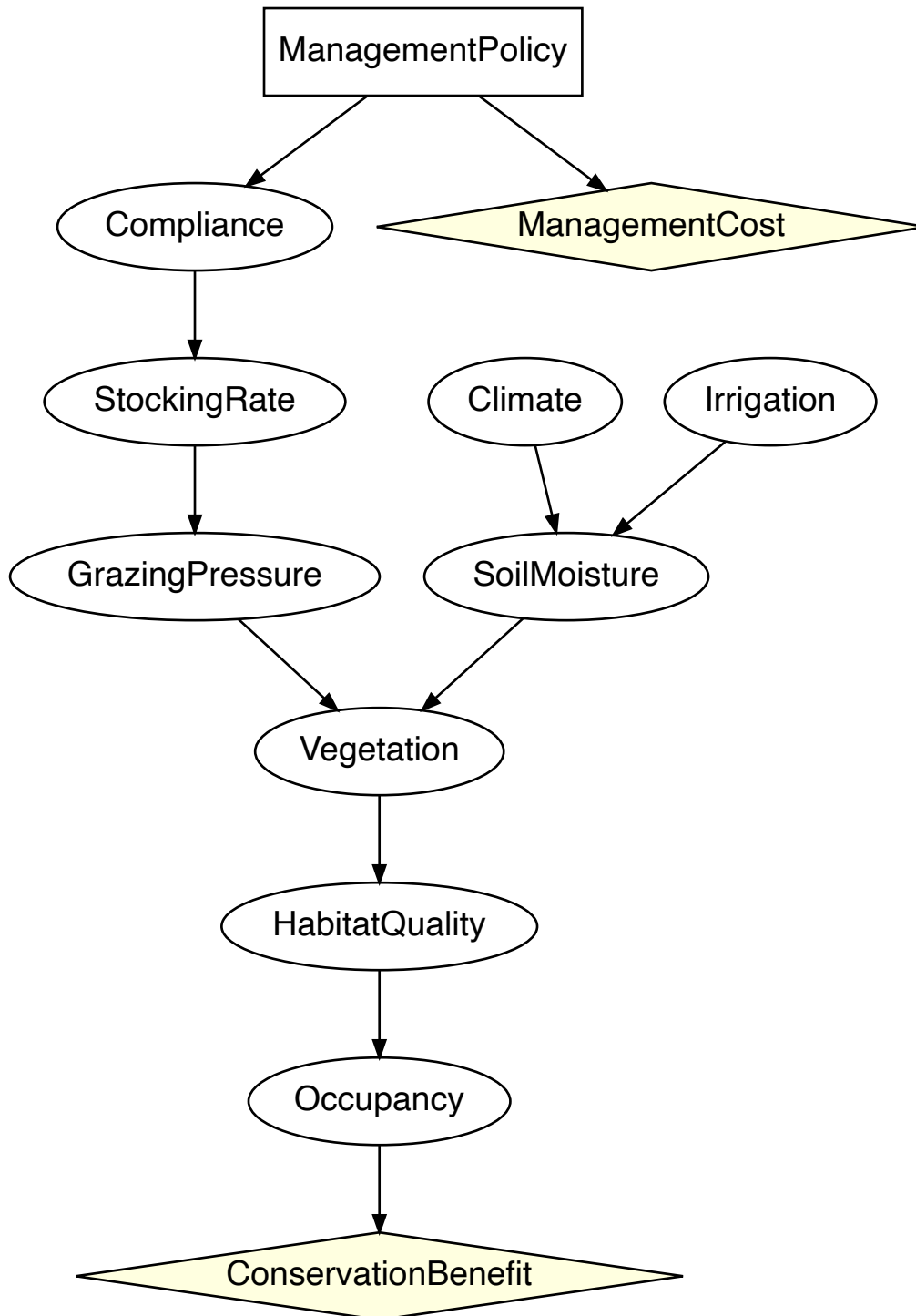
Making ManagementPolicy a decision and attaching a conservation benefit to occupancy and a cost to the restriction turns the question into a decision problem. The kernels are the ones already bound above; the diagram only adds the decision and the two utility nodes.

```
id = influence_diagram(:ManagementPolicy => [:restrict, :none], :Compliance =>
[:high, :low],
                        :StockingRate => [:low, :high], :GrazingPressure => [:low,
:high],
                        :Climate => [:dry, :normal, :wet], :Irrigation => [:low,
:high],
                        :SoilMoisture => [:low, :medium, :high],
                        :Vegetation => [:sparse, :moderate, :dense],
                        :HabitatQuality => [:poor, :good], :Occupancy => [:absent,
:present];
mechanisms = [:Compliance => :ManagementPolicy, :StockingRate
=> :Compliance,
```

```

                                :GrazingPressure ⇒ :StockingRate,
                                :SoilMoisture ⇒ (:Climate, :Irrigation),
                                :Vegetation ⇒ (:SoilMoisture,
                                :GrazingPressure),
                                :HabitatQuality ⇒ :Vegetation, :Occupancy ⇒
                                :HabitatQuality],
    decisions = [:ManagementPolicy ⇒ Symbol[]],
    utilities = [:ConservationBenefit ⇒ :Occupancy,
                :ManagementCost ⇒ :ManagementPolicy])
idm = InfluenceDiagramModel(id)
idm = bind_kernel(idm, [x ⇒ kernel(full, x)
                        for x in (:Climate, :Irrigation, :SoilMoisture, :Vegetation,
:HabitatQuality,
                                :Occupancy, :Compliance, :StockingRate,
                                :GrazingPressure)])
idm = bind_utility(idm, [:ConservationBenefit ⇒ [0.0, 1000.0], :ManagementCost ⇒
[-10.0, 0.0]])
to_graphviz(idm; states = false)

```



The expected utility of each action, and the optimum (both backends agree):

```
[a ⇒ round(expected_utility(idm, :ManagementPolicy ⇒ a); digits = 2) for a in (:restrict, :none)]
```

```
2-element Vector{Pair{Symbol, Float64}}:
 :restrict => 478.11
 :none => 465.79
```

```
sol = optimize(idm)
sol.expected_utility, policy_table(sol.strategy[:ManagementPolicy]),
optimize(idm, ExhaustivePolicySearch()).expected_utility ≈ sol.expected_utility
```

```
(478.10640718749994, fill(:restrict), true)
```

At 70% compliance the restriction is worth its cost of 10. Whether it stays worth it depends on compliance, which is exactly the point of modelling the chain: at low compliance the same restriction no longer pays.

```
println(" P(comply)   EU(restrict)   EU(none)   optimal")
for p in (0.3, 0.5, 0.7, 0.9, 1.0)
    m = bind_cpt(idm, :Compliance => [p 1-p; 0.1 0.9])
    eu = [expected_utility(m, :ManagementPolicy => a) for a in (:restrict, :none)]
    println(lpad(p, 8), " ", lpad(round(eu[1]; digits = 2), 8), " ",
            lpad(round(eu[2]; digits = 2), 8), " ", eu[1] > eu[2] ? :restrict :
            :none)
end
```

P(comply)	EU(restrict)	EU(none)	optimal
0.3	463.23	465.79	none
0.5	470.67	465.79	restrict
0.7	478.11	465.79	restrict
0.9	485.54	465.79	restrict
1.0	489.26	465.79	restrict

Finally, the value of perfect implementation: under `do(GrazingPressure = low)` the decision no longer matters for the ecology, the optimal action is the free one, and the gain over the best implementable decision is what a manager could pay for a mechanism that guaranteed low pressure.

```
direct = optimize(do_intervention(idm, :GrazingPressure => :low))
(perfect = round(direct.expected_utility; digits = 2),
 best_decision = round(sol.expected_utility; digits = 2),
 value_of_perfect_implementation = round(direct.expected_utility -
sol.expected_utility; digits = 2))
```

```
(perfect = 511.66, best_decision = 478.11, value_of_perfect_implementation = 33.55)
```

8 Summary

Writing the implementation chain out as an open network and gluing it onto the habitat network makes the difference between `do(GrazingPressure = low)` and choosing a management policy

explicit and quantitative: the implementable decision is worth strictly less, and the gap is the value of perfect implementation – what a mechanism that guaranteed compliance would be worth. Compliance, not the policy label, is the lever the numbers respond to. The next vignette, *A Song Sparrow decision network*, leaves the reference fixtures for a published network from the BNMA repository.

9 References

- Marcot, Bruce G., J. Douglas Steventon, Glenn D. Sutherland, and Robert K. McCann. 2006. “Guidelines for Developing and Updating Bayesian Belief Networks Applied to Ecological Modeling and Conservation.” *Canadian Journal of Forest Research* 36 (12): 3063–74. <https://doi.org/10.1139/x06-135>.
- Runge, Michael C., Sarah J. Converse, and James E. Lyons. 2011. “Which Uncertainty? Using Expert Elicitation and Expected Value of Information to Design an Adaptive Program.” *Biological Conservation* 144 (4): 1214–23. <https://doi.org/10.1016/j.biocon.2010.12.020>.